

SINDH MUSLIM GOVERNMENT SCIENCE COLLEGE, KARACHI
PRELIMINARY EXAMINATION PRELIM-2026

TIME : 3 HOURS

MAXIMUM MARKS 85

PHYSICS PAPER II

SECTION 'A' (Multiple Choice Questions)

Note: Attempt all questions from this section.

(17MARKS)

Q1.

- i) The absolute temperature corresponding to 212°F is:
☐ A) 273K ☒ B) 373K ☐ C) 161K ☐ D) 100K
- ii) In an isothermal process, what is the primary feature?
☒ A) Constant temperature ☐ B) No work done
☐ C) Constant volume ☐ D) No heat exchange
- iii) Generator is a device used to convert mechanical energy into
☐ A) potential energy ☒ B) electrical energy
☐ C) magnetic energy ☐ D) chemical energy
- iv) At room temperature in semi-conductors, forbidden gap between valance band and conduction band is:
☐ A) Narrow ☒ B) Wide ☐ C) not existing ☐ D) Extra wide
- v) The radiation from a black body depends upon the:
☐ A) material of the body ☐ B) shape of the body
☐ C) size of the body ☒ D) temperature of the body
- vi) Balmer series of hydrogen atom spectrum lies in
☒ A) visible region ☐ B) infrared region
☐ C) far infrared region ☐ D) ultraviolet region
- vii) The transitions of inner-shell electrons in heavy atoms give rise to
☐ A) α -rays ☐ B) β -rays ☐ C) γ -rays ☒ D) x-rays
- viii) The force between two current carrying conductors arises due to
☒ A) Magnetic effect of current ☐ B) Polarization
☐ C) Electromagnetic induction ☐ D) Electrostatic interaction
- ix) In the nuclear reaction ${}^7\text{N}^{14} + {}^2\text{He}^4 \rightarrow {}^8\text{O}^{17} + \text{-----}$, the particle in the blank will
☒ A) proton ☐ B) electron ☐ C) neutron ☐ D) gamma photon
- x) This device makes use of PN-junction;
☐ A) Wilson Cloud Chamber ☒ B) Geiger Counter
☒ C) Solid state detector ☐ D) Ionization chamber
- xi) De Broglie's wavelength of a material particle is
☒ A) h/mv ☐ B) m/hv ☐ C) v/hm ☐ D) h/mv
- xii) The device is not used in diesel engine.
☐ A) Outlet valve ☐ B) Piston ☒ C) Sparking plug ☐ D) Injector
- xiii) The primary purpose of a truth table in digital electronics is to:
☐ A) Determine the input voltage ☐ B) Measure circuit resistance
☐ C) Calculate circuit power ☒ D) Analyze output states based on inputs states
- xiv) An A.C circuit, in which capacitive reactance and inductive reactance are equal, is called:
☐ A) Reactive circuit ☐ B) Capacitive circuit
☐ C) Resistive circuit ☒ D) Resonant circuit
- xv) An elementary particle that feels all three fundamnet forces is:
☐ A) Lepton ☐ B) Quark ☒ C) Electron ☐ D) Neutrino
- xvi) Property of a body by virtue of which, it tends to regain its original shape and size, when the applied force is removed is called:
☒ A) Elasticity ☐ B) rigidity ☐ C) plasticity ☐ D) compressibility
- xvii) What is the main reason astronomers cannot directly observe black holes?
☐ A) They are too small ☐ B) They deflect light
☐ C) They are too far away ☒ D) their immense gravity traps light

$$v = f\lambda$$

$$d = \frac{v}{f}$$

PHYSICS PAPER II
SECTION 'B' (36 MARKS)

Note: Answer any **NINE** questions from this section. All questions carry equal marks.

Q2:

- (i) The binding energy of a nucleus is 225.0 MeV. What is the mass defect of the nucleus in atomic mass units?
- (ii) Calculate I_E in a transistor for which $\beta = 50$ and $I_B = 20 \mu A$.
- (iii) Calculate the root mean square speed of hydrogen at 500K (mass of proton $1.67 \times 10^{-27} \text{ kg}$ and $K = 1.38 \times 10^{-23} \text{ J/molecule} \cdot \text{K}$)
- (iv) For an inductor with an inductance (L) of 0.5 H and a frequency of 100 Hz, calculate the inductive reactance.
- (v) How fast would a rocket have to go relative to an observer for its length to be contracted to 99% of its length at rest?
- (vi) A galvanometer of resistance 60Ω gives full-scale deflection with a current of 4 mA. A resistance of 10940Ω is connected in series with a coil to convert it into voltmeter; find the range of voltmeter obtained.
- (vii) Define Compton Effect. Discuss the Compton effect qualitatively.
- (viii) Draw the circuit symbol of NOR & NAND gates and write truth table for three inputs.
- (ix) Derive formula for energy stored in an inductor.
- (x) Define rectification and describe the working of full wave **OR** half wave rectifier. **OR** Explain transistor as an amplifier.
- (xi) Define fission and fusion and briefly explain with the help of one reaction each.
- (xii) How can a galvanometer be converted in a Voltmeter **OR** an Ammeter. Derive the relevant formula.
- (xiii) Discuss pair production **OR** annihilation of matter with help of conservation laws.
- (xiv) Explain the structure of standard model, including the different types of particles. **OR** Describe in details all fundamental forces and their associated field particles.

SECTION 'C' (32 MARKS)

Note: Answer any **TWO** questions from this section.

- Q3. a) Derive an expression for the pressure of an ideal gas on the basis of kinetic molecular theory.
- b) What is an a.c generator? Write down the name of basic parts of a.c generator and explain its working with the help of diagram. Derive the formula for motional emf of an a.c generator.
- Q4. a) State postulates of Bohr's theory. Derive the formula for the n th radius of hydrogen atom.
- b) What is RLC series circuit? Draw its wave form, phasor diagram and derive the formula of impedance and phase angle of a circuit.
- Q5. a) State law of radioactive decay. Draw and explain the graph of law of radioactive decay. What is half-life of radioactive substance?
- b) Give the construction and working of Geiger counter with the diagram. Also name the other radioactive detectors. **OR** Give the construction and working of Geiger counter with the Diagram. Also name the other radioactive detector.